Daily Briefing - November 08, 2026

Your Daily Tech & Programming Digest

Sunday, November 08, 2026

1000	120960	1167	38
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

👤 Adriel Carridice 📅 2025-11-04 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Disrupted imagery, disrupted scenes: mental scene construction in behavioural-variant frontotemporal dementia

📖 1 min 📝 18 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Mohamad El Haj, Guillaume Chapelet, Dimitrios Kapogiannis, Claire Boutoleau-Bretonnière

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600483X?dgcid=rss_sd_all

Paclitaxel induces region-specific microglial remodeling associated with chronic behavioral deficits and modulated by electroacupuncture in rats

📖 1 min 📝 19 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Shinya Hiraiwa, Taiga Muramoto, Masahiro Umeda, Satomi Ebara, Fumihiko Fukuda

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004828?dgcid=rss_sd_all

Gastrin-releasing peptide in the anterior medial nucleus accumbens or paraventricular nucleus suppresses chow intake under food restriction but not palatable diet intake in nonrestricted rats

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Regan Cochran, Ella Irvin, Yiwei Tang, Stacey Chuang, Adele Nadeau, Smruti Patel, Lindsey Wilson, Weijia Zhang, Wayne E. Pratt

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004877?dgcid=rss_sd_all

SREBP1 inhibition downregulates NR2B-PSD95 interaction in glutamate excitotoxicity following cerebral ischemic reperfusion injury

📖 1 min 📝 16 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Lopmudra P. Sarode, Bhavesh Verma, Rajesh Ugale

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004841?dgcid=rss_sd_all

Individualized structural brain network and transcriptomic characterization of heterogeneity in type 2 diabetes mellitus

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Haolin Wang, Jinglei Xu, Yue Wu, Xin Du, Yang Zhang, Ying Zhai, Zhihui Zhang, Qian Wu, Minghuan Lei, Qi An, Wenjie Cai, Yanmin Peng, Feng Liu, Quan Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007207?dgcid=rss_sd_all

Correcting automatic decisions

Luis A. Mejia 2026-11-02 1 min 11 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02510-x Correcting automatic decisions

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02510-x

Misfolded tau propagates via templated seeding

Qingzhong Ren 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02512-9 Misfolded tau propagates via templated seeding

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02512-9

Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-11-02 1 min 22 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02530-7 Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02530-7

Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

Roman J. Giger 2026-11-05 1 min 21 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 05 November 2026; doi:10.1038/s41593-026-02534-3 Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02534-3

Ddx3x deficiency in mice impairs corticofugal motor circuits and causes motor deficits that are attenuated by early-life exercise

👤 Lauren DierdorffIkrame El KhaldiMarta Garcia-FornWei WangJeronimo LukinCorinne M. SmithAdele MossaZhuhao WuSilvia De Rubeisahttps://ror.org/04a9tmd77Seaver Autism Center for Research and Treatment, Icahn School of Medicine at Mount Sinai, New York, NY 10029bhttps://ror.org/04a9tmd77Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY 10029chttps://ror.org/04a9tmd77Department of Neuroscience, Icahn School of Medicine at Mount Sinai, New York, NY 10029dhttps://ror.org/04a9tmd77The Mindich Child Health and Development Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ehttps://ror.org/04a9tmd77The Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029fhttps://ror.org/04a9tmd77The Alper Center for Neural Development and Regeneration Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ghttps://ror.org/02r109517Helen & Robert Appel Alzheimer's Disease Research Institute, Feil Family Brain and Mind Research Institute, Weill Cornell Medicine, New York, NY 10021hhttps://ror.org/04a9tmd77Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, NY 10029ihttps://ror.org/04a9tmd77Department of Graduate Education, Icahn School of Medicine at Mount Sinai, New York, NY 10029

📅 2026-10-29 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAcquisition of motor skills during infancy is critical for a child's ability to interact with the physical and social environment and motor impairments are frequent in individuals with neurodevelo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535732123?af=R

Respiratory phase orchestrates human olfactory cortical dynamics

👤 Alisa Y. HathawayTodd P. Colemanahttps://ror.org/00f54p054Department of Electrical Engineering, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Department of Bioengineering, Stanford University, Stanford, CA 94305

📅 2026-11-06 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceDiagnosing smell loss in patients with cognitive disorders remains challenging because conventional tests demand complex responses that measure memory and literacy instead of basic sensory functio...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619920123?af=R

The purinergic receptor P2Y13 controls the activation and mode of division of adult ventricular–subventricular zone neural stem cells

👤 Lucía Paniagua-HerranzJulia Serrano-LópezCelia Llorente-SáezMarina Leonor Pérez-SanzDavid de Agustín-DuránPere Duart-AbadiaAna Domingo-MuelasÁfrica Vincelle-NietoPaloma BragadoDiana Manzano-FrancoMarina Arribas-BlázquezLucía GallegoAraceli Garcia-MoraLuis Alcides Olivos-OréSergio GascónÁlvaro Gutierrez-UzquizaBenedikt BerningerRaquel Pérez-SenEsmerilda G. DelicadoRosa Gómez-VillafuertesArmando Reyes-PalomaresAntonio R. ArtalejoIsabel FariñasFelipe Ortegaahttps://ror.org/02p0gd045Departamento de Bioquímica y Biología Molecular, Universidad Complutense de Madrid, Madrid 28040, SpainbInstituto Universitario de Investigación en Neuroquímica, Departamento de Bioquímica y Biología Molecular, Madrid 28040, Spainchttps://ror.org/014v12a39Instituto de Investigación Sanitaria del Hospital Clínico San Carlos de Madrid, Madrid 28040, Spaindhttps://ror.org/02p0gd045Departamento de Farmacología y Toxicología, Facultad de Veterinaria, Universidad Complutense de Madrid, Madrid 28040, Spainehttps://ror.org/043nxc105Departamento de Biología Celular y Biología Funcional, Universidad de Valencia, Valencia 46100, Spainfhttps://ror.org/043nxc105Instituto de Biotecnología y Biomedicina, Departamento de Genética, Facultad de Biología, Universidad de Valencia, Valencia 46100, Spainghttps://ror.org/02g87qh62Centro de Investigación Biomédica en Red sobre Enfermedades Neurodegenerativas, Departamento de Biología Celular y Biología Funcional, Valencia 46100, Spainhhttps://ror.org/02gfc7t72Cajal Neuroscience Center, Laboratory of direct reprogramming, Spanish National Research Council, Madrid 28805, Spainihttps://ror.org/0220mzb33Centre for Developmental Neurobiology, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomjhttps://ror.org/0220mzb33Medical Research Council Centre for Neurodevelopmental Disorders, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomkhttps://ror.org/023b0x485Institute of Physiological Chemistry, University Medical Center Johannes Gutenberg University, Mainz 55128, GermanylFocus Program Translational Neuroscience, Johannes Gutenberg University, 55128, Mainz, Germanymhttps://ror.org/036b2ww28Department of Molecular Biology and Biochemistry, Faculty of Science, University of Malaga, Malaga SE1 1UL, Spain

📅 2026-11-04 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceAdult neural stem cells (NSCs) sustain lifelong neurogenesis in the mammalian brain, yet the mechanisms controlling their activation and self-renewal remain poorly understood. We identify the puri...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2618940123?af=R

Dissociation of gravity and limb dynamics in motor control

👤 Valeriya GritsenkoTjaša KunavarMarko JamšekSergiy YakovenkoOlivier WhiteJan Babičahttps://ror.org/011vxgd24Departments of Human Performance/Neuroscience, West Virginia University, Rockefeller Neuroscience Institute, Morgantown, WV 26506bLaboratory for Neuromechanics and Biorobotics, Department of Automatics, Biocybernetics, and Robotics, Jožef Stefan Institute, Ljubljana 1000, Sloveniachttps://ror.org/02vjkv261Université Bourgogne Europe, Institut national de la santé et de la recherche médicale, Cognition, Action and Sensorimotor Plasticity, Unité Mixte de Recherche 1093, Dijon 21000, France

📅 2026-11-03 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceThe nervous system must control movement in presence of variable intrinsic and extrinsic forces, yet it has remained unclear whether mechanically equivalent forces arising from gravity and limb dy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2613283123?af=R

Lead impairs long-term memory by disrupting mitochondrial redox homeostasis in Drosophila

Cheng-Tzu HsuTe-Wei LuYa-Ting YangChun-Chao ChenYen-Hua ChiuJung-Hsuan YangZheng-Wen WuWei-Kai ChengJheng-Wei WuChen-Hao ShiuSih-Kai ChuangKuan-Lin FengHsuan-Wen LinJia-Ling YangAnn-Shyn Chiangahttps://ror.org/00zdnkx70Institute of Biotechnology, College of Life Sciences and Medicine, National Tsing-Hua University, Hsinchu 30013, Taiwanbhttps://ror.org/00mjawt10Institute of Precision Medicine, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanchttps://ror.org/00mjawt10Department of Biological Sciences, National Sun Yat-sen University, Kaohsiung 804201, Taiwandhttps://ror.org/00zdnkx70Brain Research Center, National Tsing Hua University, Hsinchu 30013, Taiwanehttps://ror.org/00mjawt10Department of Biomedical Science and Technology, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanfhttps://ror.org/00zdnkx70Institute of Systems Neuroscience, College of Life Sciences and Medicine, National Tsing Hua University, Hsinchu 30013, TaiwangGraduate Institute of Biomedical Sciences, China Medical University, Taichung 40402, TaiwanhInstitute of Molecular and Genomic Medicine, National Health Research Institutes, Miaoli 35053, Taiwanihttps://ror.org/03gk81f96Department of Biomedical Science and Environmental Biology, Kaohsiung Medical University, Kaohsiung 80780, Taiwan

📅 2026-11-04 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceEnvironmental lead exposure is strongly associated with cognitive deficits, yet how it disrupts memory at the cellular level remains unclear. Here, we show that lead selectively impairs protein-sy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2612578123?af=R

Hypothalamic lncRNA Synage prevents age-related sperm quality decline by suppressing A-type potassium currents in GnRH neurons

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📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceMen increasingly delay fatherhood, yet paternal aging lowers sperm quality and increases health risks to offspring. The mechanisms underlying these reproductive changes remain incompletely defined...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2612408123?af=R

Transcranial magnetic stimulation to the dorsolateral prefrontal cortex modulates single-neuron activity in humans

👤 Charles W. DickeyUmair HassanHiroto KawasakiAriane E. RhoneChristopher C. ClineMatthew A. HowardNicholas T. TrappAaron D. BoesJoel I. BergerCorey J. Kellerahttps://ror.org/00f54p054Department of Psychiatry and Behavioral Sciences, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Wu Tsai Neurosciences Institute, Stanford University, Stanford, CA 94305chttps://ror.org/036jqmy94Department of Neurosurgery, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Department of Psychiatry, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Department of Neurology, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Department of Pediatrics, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Iowa Neuroscience Institute, Carver College of Medicine, University of Iowa, Iowa City, IA 52242hVeterans Affairs Palo Alto Healthcare System, and the Sierra Pacific Mental Illness Research, Education, and Clinic Center, Palo Alto, CA 94305

📅 2026-11-02 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceTranscranial magnetic stimulation (TMS) to the dorsolateral prefrontal cortex is a common treatment for depression. However, it remains unknown how TMS affects single neurons in deep brain circuit...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2610803123?af=R

Hepatocyte-derived exosomal miR-122 mediates liver-to-brain crosstalk and neuroinflammation in alcohol-exposed Alzheimer’s disease mice

👤 Veronika BrezaniMarti Ortega-RiberaNeda RezagholizadehRadhika JoshiViliam BrezaniMrigya BabutaBruna de Araujo Cardoso dos SantosAdam ZivnyPrashanth Thevkar NageshIonita GhiranElizabeth Michelle ShreyerScott Brian MinchenbergAllyson FlavinGyongyi Szaboahttps://ror.org/04drvxt59Division of Gastroenterology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215bHarvard Medical School, Boston, MA 02115chttps://ror.org/04drvxt59Department of Anesthesia, Critical Care and Pain Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215dhttps://ror.org/05a0ya142Broad Institute of MIT and Harvard, Cambridge, MA 02142

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceExcessive alcohol use damages both the liver and the brain, but how these organs communicate during injury remains unclear. Our study identifies a previously unrecognized liver-to-brain communicat...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535191123?af=R

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne
2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka
2026-10-23 1 min 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1985500

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108091136&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108091136&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108091136&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108091136&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108091136&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

👤 AABC Study Group
📅 2026-11-06 📖 1 min 📝 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

👤 Xiaobing Yan
📅 2026-11-06 📖 1 min 📝 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

Junguo Duan
2026-11-07 1 min 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-11-05 1 min 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

Jing-Ning Zhu 2026-11-05 1 min 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

Yajing Meng 2026-11-05 1 min 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

> **Summary:** CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

> **Summary:** CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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A research on the typical images of animals in the dreams of Naxi.

2026-09-10 1 min 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

> **Summary:** This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

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🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

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https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

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https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

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WaveDudeNext 2D Wave Simulator

👤 sweetdingoes 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/graealex/WaveDudeNext Comments URL: https://news.ycombinator.com/item?id=50411083 Points: 32 # Comments: 8

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Agent memory needs its OpenTelemetry moment

👤 mthines 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.lorekit.io/blog/open-agent-memory-format Comments URL: https://news.ycombinator.com/item?id=50411467 Points: 81 # Comments: ...

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Llama-modchip: 284B DeepSeek on 64GB RAM, no GPU

👤 joshcsimmons 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/synchopate/llama-modchip Comments URL: https://news.ycombinator.com/item?id=50411961 Points: 59 # Comments: 24

🔗 **Read full article:**
https://github.com/synchopate/llama-modchip

JIT bitmap text rendering for microcontrollers

👤 dmit 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://linebender.org/blog/jit-bitmap-text/ Comments URL: https://news.ycombinator.com/item?id=50412072 Points: 35 # Comments: 6

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Petered (what happens after the Peter Principle)

👤 adrianhoward 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://aworkinglibrary.com/writing/petered Comments URL: https://news.ycombinator.com/item?id=50412152 Points: 79 # Comments: 37

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Person of Interest

krthr 2026-11-08 1 min 13 words HACKER NEWS

> **Summary:** Article URL: https://krthr.co/person-of-interest/ Comments URL: https://news.ycombinator.com/item?id=50412318 Points: 140 # Comments: 82

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There are 13 ways to analyse a replication study, but only one of them is coherent.

noreply@blogger.com (Daniel Lakens) 2026-05-03 6 min 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

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https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

Junguo Duan
2026-11-07 1 min 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

👤 Chunxiao Wan
📅 2026-11-05 📖 1 min 📝 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

Azfar Khalid 2026-11-05 1 min 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

2026-05-18 1 min 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

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ZapFast: A fast, native WhatsApp client in Rust and egui

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Consumer brain wave headsets threaten privacy

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Goodbye Claude, Hello Pi

👤 karagenit 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

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Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

> **Summary:** Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

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Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

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Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

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Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

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http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

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T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

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https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

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https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

📖 1 min 📝 23 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

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Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

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📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

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https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

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https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

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https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

Dynamical disorders: synchronization dynamics in complex emotional conditions

S. S. Høgenhaug 2026-11-05 1 min 207 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMental health conditions present a particular challenge to categorical nosology. We propose that such conditions are better understood as Dynamical Disorders, defined not only by symptom cluster profiles but by underlying disrupted patterns of coordination across bio-psycho-social scales...

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Epigenetic modification of neurotransmitter pathways during fasting: a systematic review

Majd A. AbuAlrob 2026-11-05 1 min 293 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundDietary restriction, encompassing intermittent fasting and caloric restriction, represents a prominent non-pharmacological approach to modulating brain aging and neuronal resilience. Emerging evidence links metabolism to long-term neural plasticity through epigenetic mechanisms, yet findin...

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Path planning algorithm based on pulse-coupled neural network with dual constraints of heading-deviation and node penalty

Xiaoyan Wang 2026-11-05 1 min 321 words FRONTIERS NEUROROBOTICS

Summary: Path planning for mobile robots is a crucial aspect of robot navigation. Traditional search algorithms are limited by heuristic design and search strategies, often struggling to balance search efficiency and path quality in intricate obstacle environments. The Pulse-Coupled Neural Network (PCNN) off...

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Research progress and prospects of cochlear implant technology

Xiang Yao 2026-11-05 1 min 251 words FRONTIERS NEUROSCIENCE

Summary: Traditional cochlear implant systems depend on external devices and face challenges related to convenience, esthetics, and auditory performance in complex soundscapes. Recently, research has shifted from “functional substitution” to “functional enhancement” and “biological fusion.” However, most eme...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

👤 Peng-Hsiang Hung
📅 2026-11-05 📖 1 min 📝 61 words

LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho
📅 2026-11-05 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

> **Summary:** [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

> **Summary:** CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

👤 Lara Z Maliske
📅 2026-11-05 📖 1 min 📝 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell
📅 2026-11-05 📖 1 min 📝 55 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Giving back to the community - The Complete Backend Development Course

/u/ OrganizationGreat823 2026-11-07 1 min 78 words REDDIT PYTHON

Summary: <!-- SC_OFF --> Hey everyone, I decided to make my course free in order to help people.
 This course is my backend development course which is about SQL, Python, APIs, Docker, Kubernetes, Linux, Git & More The link is: <a href="https://www.youtube.com/watch?v=CBIu6h...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xqgnoh/giving_back_to_the_community_the_complete_backend/

Sunday Daily Thread: What's everyone working on this week?

/u/ AutoModerator 2026-11-08 1 min 211 words REDDIT PYTHON

Summary: <!-- SC_OFF --> <h1>Weekly Thread: What's Everyone Working On This Week? 🛠️</h1> Hello r/Python! It's time to share what you've been working on! Whether it's a work-in-progress, a completed masterpiece, or just a rough idea, let us know what you're up to!</p...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xqknnl/sunday_daily_thread_whats_everyone_working_on/

Claude Code's "Don't Ask Again" exposes passwords in 1,122 public GitHub files

👤 shmulc 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://apiiro.com/blog/coding-agents-leaking-secrets-github-dont-ask-again Comments URL: <a href="https://news.ycombinator.com/item?id=50407882">https://news.ycombinator.com/item?id=5040...

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https://apiiro.com/blog/coding-agents-leaking-secrets-github-dont-ask-again

Shadow of the Sun (2021)

👤 andsoitis 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mattlakeman.org/2021/07/27/shadow-of-the-sun/ Comments URL: https://news.ycombinator.com/item?id=50408104 Points: 18 # Comments...

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MeccanicOS

👤 tomcam 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://meccanicos.com/ Comments URL: https://news.ycombinator.com/item?id=50408129 Points: 14 # Comments: 4

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https://meccanicos.com/

Show HN: An offline desktop companion to help maintain a healthy blink rhythm

👤 prnysoni 📅 2026-11-07 📖 1 min 📝 212 words HACKER NEWS

> **Summary:** Profession like software engineer, mechanical / graphics designer takes a lot of strain on your eyes, observing minute details for hours and hours on screen can significantly reduce the blink rate which you might not even notice until it gets irreversibly painful, leaving you with dry eyes. I am ...

🔗 **Read full article:**
https://eyesrhythm.com/

Making any integer with four 2s (2025)

👤 germanrabbit 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://eli.thegreenplace.net/2025/making-any-integer-with-four-2s/ Comments URL: https://news.ycombinator.com/item?id=50408713 ...

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IRA Velinsky – Commodore's Industrial Designer

👤 erickhill 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.commodore.ca/commodore-history/ira-velinsky-commodores-amazing-industrial-designer/ Comments URL: <a href="https://news.ycombinator.com/item?id=50408750">h...

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GitHub New pull request overview page in public preview

👤 xiaoyu2006 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.blog/changelog/2026-11-05-new-pull-request-overview-page-in-public-preview/ Comments URL: <a href="https://news.ycombinator.com/item?id=50408931">https://new...

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Commodore PET – The Worlds First Personal Computer

👤 andsoitis 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.commodore.ca/commodore-products/commodore-pet-the-worlds-first-personal-computer/ Comments URL: <a href="https://news.ycombinator.com/item?id=50408961">https...

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Type S and M errors as a “rhetorical tool”

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-09-28 📖 17 min 📝 3572 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-10-29 📖 27 min 📝 5516 words

TWENTY PERCENT STATISTICIAN

> **Summary:** In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

> **Summary:** A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Principles of coarse-scale functional organization in occipitotemporal cortex

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11319-0

Corticosteroid treatment dysregulates perirhinal circadian rhythms and synaptic function in rats

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11316-3

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

Bite-block effects on native s/ʃ and English r/l in child speakers of a language with/without this contrast

F. Sayako Earle 2026-11-02 1 min 171 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Learning to produce non-native speech contrasts requires developing new phonological categories as well as new articulatory-perceptual mappings for these new categories. However, within the same speakers, it remains unclear how these new categories compare to contrasts that are already fully establi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949253

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao 2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261107160441&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261107160441&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

Peng-Hsiang Hung 2026-11-05 1 min 61 words LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho
📅 2026-11-05 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

👤 AABC Study Group
📅 2026-11-06 📖 1 min 📝 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

👤 Xiaobing Yan
📅 2026-11-06 📖 1 min 📝 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

> **Summary:** [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

> **Summary:** CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

Heather M Joseph
2026-11-04 1 min 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

Lara Z Maliske
2026-11-05 1 min 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell 📅 2026-11-05 📖 1 min 📝 55 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Implicit learning of melodic structure: A role for pitch?

📅 2024-01-22 📖 1 min 📝 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

Summary: I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Mysterious Lazy Imports

👤 /u/__peg_parser__ 📅 2026-11-07 📖 1 min 📝 86 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Run this on Python 3.15: <pre><code>>>> lazy import json >>> type(globals()['json']) <class 'lazy_import'> </code></pre> Now try separating that into an innocent variable: <pre><code>>>> lazy import json >>> x = g...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xqblqi/mysterious_lazy_imports/

Every processor die anyone has photographed and given away

👤 user_7832 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://sheets.works/data-viz/cpu-dies Comments URL: https://news.ycombinator.com/item?id=50405782 Points: 47 # Comments: 11

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MaxxBench – "Nice DeepSWE score. But is it true?"

👤 antiuniverse 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://maxxbench.unlap.ai/research/why-maxxbench/ Comments URL: https://news.ycombinator.com/item?id=50406222 Points: 106 # Comments: 26<...

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Show HN: Coldframe, a honeypot that poses as a supermarket fridge controller

👤 hedgerow-dev 📅 2026-11-07 📖 1 min 📝 258 words HACKER NEWS

> **Summary:** I built Coldframe, a honeypot that pretends to be a supermarket refrigeration controller. Inspired by the events on industrial fridge installations in the past weeks, I wanted to see what people would try against a convincing fake. To learn more about this I built a honeypot that resembled perf...

🔗 **Read full article:**
https://github.com/hedgerow-dev/coldframe

Five memorable books about programming (2008)

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> **Summary:** Article URL: https://prog21.dadgum.com/19.html Comments URL: https://news.ycombinator.com/item?id=50406319 Points: 45 # Comments: 20

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GeoWorks, Microsoft Windows' Upstart '90s Competitor

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> **Summary:** Article URL: https://www.atlasobscura.com/articles/the-history-of-geoworks-microsoft-windows-upstart-90s-competitor Comments URL: <a href="https://news.ycombinator.com/item?...

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The end of Microsoft Research's AI Frontiers group

👤 srean 📅 2026-11-07 📖 1 min 📝 16 words HACKER NEWS

> **Summary:** https://archive.is/AaY5O <a href="https://businessexaminer.ca/victoria-articles/item/microsoft-researcher-john-langford-exits-as-ai-frontiers-group-is-declared/" rel="nofollow">https://businessexaminer.ca/victoria-articles/item/microsoft...<...

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What I run on my $10 USD OpenStick homelab

👤 LaurensBER 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.bosscher.ch/posts/openstick/ Comments URL: https://news.ycombinator.com/item?id=50406670 Points: 138 # Comments: 25

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https://blog.bosscher.ch/posts/openstick/

Automatic computational reproducibility checks with Metacheck

noreply@blogger.com (Daniel Lakens)
2026-10-30 19 min 3854 words

TWENTY PERCENT STATISTICIAN

Summary: When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

Steve Luck
2024-03-16 1 min 106 words

ERP BOOT CAMP

Summary: The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

> **Summary:** The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Dynamic muscle proprioception impairment in adolescent idiopathic scoliosis revealed by tendon vibration

📖 1 min 📝 29 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Daniela M Pinzon-Corredor, Christophe Boulay, Alexis Ulian, Guillaume Authier, Nicole Levy, Fabienne Levy, Jean-Luc Jouve, Corentin Petitpas, Sébastien Pesenti, Anne Kavounoudias

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Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

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https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Brain-heart decoupling in somatic depression: Desynchronization between salience network neuropsychological connectivity and heart rate variability

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yi Xia, Junling Sheng, Xiaoqin Wang, Ciqing Bao, Yujue Wang, Xumiao Wang, Zhijian Yao, Qing Lu

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https://www.sciencedirect.com/science/article/pii/S1053811926006464?dgcid=rss_sd_all

Do autistic traits modulate the neural processing of logical rule violations in visual and auditory modalities?

📖 1 min 📝 28 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Junyu Wu, Rong Huang, Jingyang Yu, Jiayi Shi, Tianze Wang, Yuting Zhang, Yutong Chen, Jie Li, Shubo Shui, Jing Meng

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006427?dgcid=rss_sd_all

Feasible closed-loop stimulation for neural latent dynamics control

👤 Johnsen, K. A., Lai, Y., Vivekanandasarma, L., Berman, G. J., Pandarinath, C., Rozell, C. J.

📅 2026-11-07 📖 1 min 📝 257 words BIORXIV NEUROSCIENCE

Summary: Modern neuroscience benefits from powerful computational and experimental methods for studying neural populations, including latent dynamics models that predict behavior with few dimensions. But despite their explanatory utility, latent factors are seldom targeted explicitly with experimental pertur...

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https://www.biorxiv.org/content/10.64898/2026.11.06.762216v1?rss=1

Identification of Kinome Biomarkers and Therapeutic Rescue in cGMP-Linked Retinal Degeneration

Topcu Okan, I. C., Ahmadian, M., Altay, H. Y., Pal Singh, S., Kheirandish, G., Ozer, S., Gaillard, P., Bouffard, R., Neutzner, A., Esmen, K., Tschopp, M., Lawrence, M., Diril, M. K., Paquet-Durand, F., Agca, C.

2026-11-07 1 min 195 words BIORXIV NEUROSCIENCE

Summary: Inherited retinal degenerations, such as Retinitis Pigmentosa (RP) and Leber's congenital amaurosis (LCA), are major causes of progressive and irreversible vision loss. The large number of genes associated with these disorders makes their clinical and molecular characterization difficult, while syst...

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Epidermal Langerhans Cells Act as Neuroimmune Regulators of Pain and Itch

Sakai, K., Ishida, H., Pavlenko, D., Suehiro, M., Oshea, T., Dorning, J., Sanders, K., Sanchez, C., Lopez, A., Talamas, S., Boulina, M., Griswold, A. J., Akiyama, T.

2026-11-07 1 min 158 words BIORXIV NEUROSCIENCE

Summary: Neuroimmune interactions in the skin coordinate immune defense and somatosensory perception, yet how immune cells shape sensory signaling remains poorly defined. Here, we show that Langerhans cells (LCs), epidermal antigen-presenting cells, function as local immune-derived amplifiers of cutaneous se...

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Beyond performance scores: directed functional connectivity as a brain-based biomarker for tracking motor skill learning and retention

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Longitudinal prediction of biological aging: replication and extension of DunedinPACNI

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https://www.nature.com/articles/s41514-026-00562-8

Insular motor and error encoding supports self-correcting intracranial brain-computer interface

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🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78804-2

Single-neuron encoding of rapidly learned visual information reshapes human perception

Gabriel Kreiman 2026-11-04 1 min 38 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

2026-11-04 1 min 54 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

Editorial: Data-driven brain imaging and signal analysis: methods from acquisition to clinical insight

Zhiliang Wei 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2010682

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

Peng-Hsiang Hung 2026-11-05 1 min 61 words LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho 📅 2026-11-05 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-11-05 1 min 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

Jing-Ning Zhu 2026-11-05 1 min 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

Lara Z Maliske
2026-11-05 1 min 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

👤 Chunxiao Wan
📅 2026-11-05 📖 1 min 📝 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan
📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

Azfar Khalid 2026-11-05 1 min 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell
📅 2026-11-05 📖 1 min 📝 55 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

📅 2024-01-22 📖 1 min 📝 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Python backend support needed.

👤 /u/Strange-Flatworm-742 📅 2026-11-07 📖 1 min 📝 57 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I have one task and i am looking for someone who can work tomorrow for that 1 task , python backend engineer who has good experience will be nice. #python #senior engineer #backendengineer #worksupport #freelancer <!-- SC_ON -->   submitted by ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xq8kxq/python_backend_support_needed/

the faulthandler i turned on to catch crashes was causing them (windows, all_threads=True)

👤 /u/Stunning-Sherbet1853 📅 2026-11-07 📖 1 min 📝 399 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> i run a long-lived python 3.11 backend on windows 11. at the end of october it died mid-task once with nothing in the logs, so i turned on faulthandler.enable(file=..., all_threads=True) to see where it died next time. over the next three days it died four ti...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xq4rqx/the_faulthandler_i_turned_on_to_catch_crashes_was/

Running a law firm on Free software

👤 DeepLogin 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://decoded.legal/blog/2026/11/running-a-law-firm-on-free-software-2026-edition/ Comments URL: <a href="https://news.ycombinator.com/item?id=50403066">https://news.ycombinato...

🔗 **Read full article:**
https://decoded.legal/blog/2026/11/running-a-law-firm-on-free-software-2026-edition/

Every Mac comes with an (undocumented) family member classifier

👤 infwhispers 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://8rank.ai/game/apple-intelligence-personal-knowledge-graph Comments URL: https://news.ycombinator.com/item?id=50403097 Poin...

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https://8rank.ai/game/apple-intelligence-personal-knowledge-graph

Upload Your RAM

👤 susanthenerd 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://uploadyourram.org Comments URL: https://news.ycombinator.com/item?id=50403214 Points: 249 # Comments: 232

🔗 **Read full article:**
https://uploadyourram.org

The $8 Linux Computer (2022)

👤 zenincognito 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: http://thelittleengineerthatcould.blogspot.com/2022/12/the-8-linux-computer.html Comments URL: <a href="https://news.ycombinator.com/item?id=50404599">https://news.ycombinator.com/it...

🔗 **Read full article:**
http://thelittleengineerthatcould.blogspot.com/2022/12/the-8-linux-computer.html

The Gordon Gekko Effect: The Role of Culture in the Financial Industry [pdf]

👤 tylerdane 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.newyorkfed.org/medialibrary/media/research/epr/2016/epr_2016_gordon-gekko-effect_lo.pdf Comments URL: <a href="https://news.ycombinator.com/item?id=504...

🔗 **Read full article:**
https://www.newyorkfed.org/medialibrary/media/research/epr/2016/epr_2016_gordon-gekko-effect_lo.pdf

Ask HN: How did you convince family and friends to join your XMPP/Matrix server?

👤 mtlynch 📅 2026-11-07 📖 1 min 📝 280 words HACKER NEWS

> **Summary:** I love open-source software and open protocols, but my family and friends primarily communicate through SMS for exactly the reasons in https://xkcd.com/2365/ Whenever XMPP or Matrix threads pop up on HN, several commenters casually mention their "friends and family XMPP (or Matrix) server." H...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50405250

Show HN: We compressed a 31B video+audio diffusion model to 14B, 6 steps, INT8

👤 jiant 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/Video-Rebirth/hyperflow-h3-flash Comments URL: https://news.ycombinator.com/item?id=50405263 Points: 3 # Comments: 0</...

🔗 **Read full article:**
https://github.com/Video-Rebirth/hyperflow-h3-flash

They don't make 'em like that any more: Borland Turbo Pascal 7

👤 andsoitis 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://kevinboone.me/tpwin.html Comments URL: https://news.ycombinator.com/item?id=50405678 Points: 49 # Comments: 31

🔗 **Read full article:**
https://kevinboone.me/tpwin.html

How Long Can Life Wait?

👤 Ariarule 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mceglowski.substack.com/p/how-long-can-life-wait Comments URL: https://news.ycombinator.com/item?id=50405789 Points: 24 # Co...

🔗 **Read full article:**
https://mceglowski.substack.com/p/how-long-can-life-wait

Apple Keyboard has a blacklist of phrases you can't swipe

👤 user_7832 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.theguardian.com/technology/2026/nov/06/struggling-to-spell-male-chauvinist-pig-apple-grey-list-means-your-iphone-wont-help-you">https://www.theguardian.com/technology/2026/nov/06/struggling-to-spell-male-chauvinist-pig-apple-grey-list-means-your-iphone-wont-help-...

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https://www.theguardian.com/technology/2026/nov/06/struggling-to-spell-male-chauvinist-pig-apple-grey-list-means-your-iphone-wont-help-you

Charitable giving with earned and windfall money: behavioral and ERP responses to donation-risk information

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jing Wang, Xiaoya Li, Siqi Jian, Peng Li

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004750?dgcid=rss_sd_all

Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Fan Yu, Yi Feng, Yun Miao, Guangxu Xu

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https://www.sciencedirect.com/science/article/pii/S0006899326004671?dgcid=rss_sd_all

TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

📖 1 min 📝 14 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004804?dgcid=rss_sd_all

Cannabidiol’s prosexual effects in sexually active and sexually inhibited male rats

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Gabriela Rodríguez-Manzo, Ana Canseco-Alba, Jorge Rojas-Hernández

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006998?dgcid=rss_sd_all

Machine learning of connectome-wide functional connectivity in spinal muscular atrophy

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Yufen Li, Huasong Cai, Mengyu Zhang, Mengzhen Yan, Shasha Lan, Huirong Nie, Wanqing Shen, Cui Yan, Boyan Xu, Yujian Liang, Wen Tang, Yijuan Li, Pei Xiang, Yingqian Chen

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007037?dgcid=rss_sd_all

Spectral-spatial modeling of neural couplings in functional connectivity of motor networks

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Hamidreza Abbaspour, Melissa H. Wade, Christopher P. Hurt, Christopher L. Gonzalez, Harrison C. Walker, Zachary T. Irwin

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006294?dgcid=rss_sd_all

Dual kinase inhibition provides early clues to restoration of neuroimmune homeostasis in an AAV-haSyn model of Parkinsons disease

👤 Chiang, W., Sandoval, I. M., Li, H., Bernal-Conde, L. D., Squire, R., Washburn, M. K., Schwartz, J. L., Squire, E., Wagner, E. J., Manfredsson, F. P., Gelbard, H. A.

📅 2026-11-07 📖 2 min 📝 409 words BIORXIV NEUROSCIENCE

> **Summary:** Background Previous studies of drugs targeting mixed linage kinase type 3 (MLK3) and/or leucine rich repeat kinase type 2 (LRRK2) in sporadic Parkinsons disease (PD) have failed to meet clinical endpoints despite multiple lines of evidence implicating these kinases in pathogenesis. This has reduced ...

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https://www.biorxiv.org/content/10.64898/2026.11.06.764938v1?rss=1

Shikonin alleviates bone cancer pain through suppression of spinal IGF2-associated p38 MAPK signaling

Ma, z., Jia, J., Lu, D., Wanying, Y., Xiaogeng, H., Shuyao, Z., Xinfang, F., Jing, Z., Fan, B., Longsheng, X., Ming, Y.

2026-11-07 1 min 361 words BIORXIV NEUROSCIENCE

Summary: Background Bone cancer pain (BCP) is a debilitating complication of advanced cancer that substantially impairs physical function and emotional wellbeing. Current pharmacological treatments often provide incomplete analgesia and may be limited by adverse effects. Shikonin, a bioactive naphthoquinone ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.06.764749v1?rss=1

Circulating branched-chain amino acids and multimodal signatures of incident Parkinson’s disease: a prospective cohort study

2026-11-07 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41531-026-01633-z

Cognitive impairment is associated with lower extremity function among community-dwelling older adults in Taiwan

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-74879-5

Impaired spatial memory destabilization and updating in middle-aged male mice is reversed by muscarinic receptor stimulation

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79144-3

Virtual screening of Withania somnifera phytochemicals identifies scopoletin and linoleic acid as predicted dual-target AChE/BACE1 candidates for alzheimer’s disease

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-71874-8

Electroencephalographic patterns of equine sleep: comparative description and neurophysiological implications of recumbent and standing sleep

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-73425-7

Virus induced cochlear inflammation alters stria vascularis function and auditory development

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78638-4

Clinical classification of multiple sclerosis by machine learning

📅 2026-11-07 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79792-5

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

👤 Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Editorial: Exploring the role of criticality in neural systems

Tiago L. Ribeiro 2026-11-04 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco
📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

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Editorial: Women in speech and language 2025

👤 Anastasia M. Raymer
📅 2026-11-04 📖 1 min 📝 0 words

FRONTIERS HUMAN NEUROSCIENCE

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Music with monaural beats for anxiety reduction and mood enhancement in individuals with moderate trait-anxiety

Gianluca Micchi 2026-11-04 1 min 279 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Monaural beats embedded in music are a popular anxiety-relief intervention, yet implementation parameters and efficacy remain poorly understood. Across two studies of online participants with moderate trait anxiety, we tested whether theta-frequency (4–7 Hz) monaural beats in music produce anxiolyti...

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The association between dietary protein intake and mental disorders in older adults: a scoping review

Aliza Haslinda Hamirudin 2026-11-04 1 min 280 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMental disorders are increasingly prevalent among older adults, significantly affecting their quality of life and overall wellbeing. Recent evidence suggests that dietary factors, particularly protein intake, may play a role in mental health outcomes. However, the association between dieta...

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Editorial: Understanding consciousness: neural, cognitive, and philosophical perspectives

👤 Joachim Keppler 📅 2026-11-04 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

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Editorial: BCIs for monitoring and treating mental health conditions

👤 Andrei Dragomir 📅 2026-11-04 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

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Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng 2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

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Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

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MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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Retraction: Hypothermia impairs glymphatic drainage in traumatic brain injury as assessed by dynamic contrast-enhanced MRI with intrathecal contrast

2026-11-04 1 min 0 words FRONTIERS NEUROSCIENCE

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The butyrate-m6A-BDNF axis: a hypothesis of gut-brain signal transduction in PSD

Dajun Gu 2026-11-04 1 min 151 words FRONTIERS NEUROSCIENCE

Summary: A reduction in brain-derived neurotrophic factor (BDNF) protein levels has been reported in post-stroke depression (PSD). However, whether transcript abundance and translation efficiency change in parallel or become uncoupled in PSD remains to be determined. In addition to classical transcriptional ...

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TEMPO-taVNS: a multimodal platform optimizing thermal, electrical, and mechanical auricular stimulation

Matthew P. Ward 2026-11-04 1 min 176 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is typically delivered electrically, while alternative stimulation modalities and their potential for personalization remain underexplored. We present TEMPO-taVNS (Thermal, Electrical, and Mechanical Parameter Optimization for taVNS), a human-...

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Baseline cognition accounts for most predictive information on cognitive recovery during rehabilitation after carbon monoxide poisoning

Peihui Liu 2026-11-04 1 min 271 words FRONTIERS NEUROSCIENCE

Summary: BackgroundCognitive change during rehabilitation after carbon monoxide (CO) poisoning may differ from prognosis at acute presentation. We examined whether information recorded in a rehabilitation pathway added predictive information beyond baseline cognition.MethodsThis single-center retrospective c...

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A purinergic receptor dominance model for microglial state transitions: balancing perturbation and restraint during neuroinflammation

Minke Tang 2026-11-04 1 min 183 words FRONTIERS NEUROSCIENCE

Summary: Microglia show substantial functional heterogeneity during neuroinflammatory responses, yet the upstream mechanisms through which they interpret extracellular perturbation-associated signals and transition between functional states remain incompletely understood. Purinergic signaling is well positio...

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Delineating neural contributions to electroencephalogram-based speech decoding

Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

2026-10-30 1 min 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min 241 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

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NeuroTest: a benchtop testbed for evaluating sensing-capable electrophysiology and neurostimulation systems

Morrow JK, H Cho, M Benjaber, Denison TJ, Widge AS and Herron JA

2026-11-04 1 min

247 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. The development of increasingly complex implantable neuromodulation devices requires extensive benchtop validation and testing prior to in vivo implementation. Our goal is to enable this in vitro testing by creating a benchtop system that is accessible, easy-to-use, and functions reliably...

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Towards semantic brain-computer interfacing: the graded inventory of semantic triggers (GIST) paradigm

Joshua James Podmore, Moein Radman, Riccardo Poli, Silke Paulmann and Ian Daly

2026-11-04

1 min 299 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Semantic brain-computer interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual p...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

👤 Peng-Hsiang Hung
📅 2026-11-05 📖 1 min 📝 61 words

LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho
📅 2026-11-05 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

Heather M Joseph
2026-11-04 1 min 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

Lara Z Maliske
2026-11-05 1 min 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell
📅 2026-11-05 📖 1 min 📝 55 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

📅 2026-07-30 📖 1 min 📝 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001096

Internals of Asyncio

👤 /u/humbulani-ndou 📅 2026-11-07 📖 1 min 📝 55 words REDDIT PYTHON

Summary: <!-- SC_OFF --> This article is a deep dive into the internals of the asyncio Python standard library. It is written from a debugging perspective, stepping through the code line by line. Article: <a href="https://origintracer.app/excerpt/asyncio-internals/">internals of asyn...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xq2x06/internals_of_asyncio/

New software package: ERPLAB Studio

Steve Luck 2024-06-12 2 min 449 words ERP BOOT CAMP

Summary: We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Recording and slides now available for ERPLAB Studio webinar

Steve Luck 2024-06-28 1 min 30 words ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

> **Summary:** In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

> **Summary:** IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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http://ieeexplore.ieee.org/document/11454471

Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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http://ieeexplore.ieee.org/document/11364153

Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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http://ieeexplore.ieee.org/document/11359456

Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11741003

Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11741004

Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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http://ieeexplore.ieee.org/document/11741002

Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11743154

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11740941

Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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http://ieeexplore.ieee.org/document/11740999

The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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http://ieeexplore.ieee.org/document/11741000

Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

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https://www.embs.org/uncategorized/welcoming-the-newly-elected-embs-adcom-members-for-2026-2028/

Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

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EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

> **Summary:** Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

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https://www.embs.org/featured-news/ieee-pulse-editor-in-chief-search-submit-and-application/

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Compositional computations in the primate frontal cortex

👤 Tang, C., Jazayeri, M. 📅 2026-11-06 📖 1 min 📝 188 words BIORXIV NEUROSCIENCE

> **Summary:** A hallmark of intelligence is the ability to apply familiar operations to new inputs without relearning. How neural circuits carry a learned computation into states never visited during training remains unclear. Here we combined neural recordings, circuit modeling and human experiments to investigat...

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https://www.biorxiv.org/content/10.64898/2026.11.05.764642v1?rss=1

Increased Cognitive Load Expands, Stabilizes, and Aligns Visual Representation Across the Human Brain

Qu, S., Chun, M. M., Xu, Y.
2026-11-06 1 min 157 words BIORXIV NEUROSCIENCE

Summary: Cognitive load refers to the mental effort required to perform a task. While previous research has extensively examined its detrimental effects on neural and behavioral performance, much less is known about how the brain actively adapts to heightened demands. Across two fMRI experiments involving vi...

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Confidence: Its Measurement, Relation to Behavior, and Neural Representation

Tsutsumi, M., Ogawa, T., Kishi, T., Kawanabe, M., Kuroda, T.
2026-11-06 1 min 150 words BIORXIV NEUROSCIENCE

Summary: Confidence (certainty/uncertainty about one's own decision) has attracted considerable attention in psychology, neuroscience, and artificial intelligence. Yet its relation to behavior remains unclear: Some treat confidence as a cause of behavior whereas others view it as a state controlled by enviro...

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Deprivation of dietary fiber promotes low-grade neuroinflammation and anxiety-like behavior in mice

Oyetunde, A., Pyaram, D. N., Ngo, V., Li, C., Abo, H., Zou, J., Mabb, A. M., Gewirtz, A. T.

2026-11-06 1 min 245 words BIORXIV NEUROSCIENCE

Summary: Background: Intestinal health is an important determinant of brain function in that an array of gut perturbants, especially those targeting microbiota, influence behavior. Fermentable dietary fiber nourishes gut microbiota and, consequently, promotes gut health. We hypothesized that depriving mice o...

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Phase-locked dual-band vibration bursts strengthen anti-kindling by vibrotactile coordinated reset stimulation in a spiking network model

Patki, S. A., SKM, V.

2026-11-06 1 min 276 words BIORXIV NEUROSCIENCE

Summary: Parkinsons disease and related movement disorders are accompanied by pathologically synchronized neuronal activity in cortico-basal ganglia circuits. Coordinated reset (CR) stimulation was derived computationally as a means of suppressing such activity at its source, by resetting the phases of the s...

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Seed Amplification Assay-Based Detection of Alpha-Synuclein Aggregates Outperforms Immunohistochemical Analysis in the Intestinal Mucosal Biopsies of Parkinson's Disease and Multiple System Atrophy Participants

👤 Keshavarzian, A., Engen, P. A., Soto, C., Voigt, R. M., Killinger, B., Shaikh, M., Pinho, M. A., Pritzkow, S., Duvernay, J. B., Barnett, M. K., Green, S. J., Abdel-Mohsen, M., Sharma, A., Goetz, C. G.

📅 2026-11-06 📖 1 min 📝 253 words BIORXIV NEUROSCIENCE

Summary: Objective: To determine whether an -synuclein (-Syn) seed amplification assay (SAA) detects seeding-competent aggregates in endoscopic mucosal biopsies, differs by gastrointestinal region, registers pathology more frequently than immunohistochemistry (IHC), and associates with Parkinson disease (PD)...

🔗 **Read full article:**
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Concussion decreases brain signal variability

Schultz, D. H., Bouchard, H. C., Barnwell, B. C., Muncy, N. M., Wilcox, R. R., Savage, C. R., Neta, M., Barbey, A. K.

2026-11-06 1 min 236 words BIORXIV NEUROSCIENCE

Summary: Sports-related concussion represents a major public health concern, with millions of cases diagnosed annually. Despite significant methodological advances in functional brain imaging, a clinically meaningful neuroimaging biomarker of concussion has yet to be established. This may be due in part to a...

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Neurons entrain and shape an intrinsic arteriole rhythm that distributes blood across the cortical mantle

Duckworth, J., Broggini, T., Chhabria, K., Neill, P., Sancho, M., Sellier-Prono, M., Iyengar, S., Vergassola, M., Nelson, M. T., Kleinfeld, D.

2026-11-06 1 min 146 words BIORXIV NEUROSCIENCE

Summary: The cerebral arteriole network distributes blood across cortex through branches whose diameters oscillate near a frequency fvaso ~ 0.1 Hz. We asked whether the spatiotemporal pattern of these vaso-oscillations reflects neuronal entrainment of an intrinsic vascular oscillation. Three findings support...

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SNAzzy: an image processing pipeline for investigating global Synchronous Network Activity

Paiva, C. D., Evora, A. J., Zheng, S., Carreira-Rosario, A.

2026-11-06 1 min 193 words

BIORXIV NEUROSCIENCE

Summary: Genetically encoded fluorescent indicators are powerful tools for monitoring biological processes in live samples (Lin and Schnitzer 2016; Nakai, Ohkura, and Imoto 2001). When combined with a large field of view, a single time-lapse recording has the potential to capture many specimens, facilitating...

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Probing EEG Foundation Model Attention Maps for Lightweight Interpretable Emotion Recognition

Arvanitopoulos, L. N., Dragomir, A., Fidas, C. A., Bezerianos, A.

2026-11-06 1 min 256 words

BIORXIV NEUROSCIENCE

Summary: Foundation Models (FMs) trained on EEG signals promise subject- and dataset-independent pipelines for brain-computer interface (BCI) applications, but have yet to consistently outperform task-specific architectures, and their large representations are costly to compute and store. As a means of advan...

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Hippocampal transformations occur along dimensions of memory interference

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Media-based nature exposure and negative psychological states: effects of natural environment videos on attentional bias and valence perception

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Effects of maximum teleportation distance on wayfinding in virtual reality: the role of optical flow cues

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Propofol induces sustained suppression of alcohol drinking via habenular activation in mice

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https://www.nature.com/articles/s41467-026-79118-z

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

👤 Peng-Hsiang Hung 📅 2026-11-05 📖 1 min 📝 61 words LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho 📅 2026-11-05 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

Lara Z Maliske
2026-11-05 1 min 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

Yajing Meng 2026-11-05 1 min 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell 📅 2026-11-05 📖 1 min 📝 55 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Psychomusicology: A resounding closing cadence.

📅 2024-01-22 📖 1 min 📝 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

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http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual's cognition in solving a task. However, very little is known about how an individual's conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

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http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

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http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

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http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

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http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

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https://fmhy.net/posts/may-2026

Monthly Updates [November]

📅 2026-11-01 📖 5 min 📝 1013 words FMHY

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🔗 **Read full article:**
https://fmhy.net/posts/nov-2026

VSCODE Autofill

👤 /u/18ahmed 📅 2026-11-06 📖 1 min 📝 143 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hi lads, Just got back into programming as my job didn’t need it until now. I’m practising writing python scripts to automate tasks for me, however iv noticed that VScode autofills parts out. I do have an idea of what I want and start it off, VSCODE intellige...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xprvie/vscode_autofill/

Saturday Daily Thread: Resource Request and Sharing! Daily Thread

👤 /u/AutoModerator 📅 2026-11-07 📖 1 min 📝 180 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Resource Request and Sharing 📚</h1> Stumbled upon a useful Python resource? Or are you looking for a guide on a specific topic? Welcome to the Resource Request and Sharing thread! <h2>How it Works:</h2> Request: Can't...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xps4q4/saturday_daily_thread_resource_request_and/

Firecracker VM Escape: PwnEscape Part 2

👤 SerCe 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://pwn.ai/blog/firecracker-escape Comments URL: https://news.ycombinator.com/item?id=50397219 Points: 196 # Comments: 96

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A Jev that can ask Opus for help gives Opus accuracy at 4x lower cost

👤 theanomaly 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

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Google has acquired Crusoe's data center site in southeast Wyoming

👤 rexbee 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.crusoe.ai/resources/newsroom/crusoe-wyoming-data-center-campus-update Comments URL: <a href="https://news.ycombinator.com/item?id=50397711">https://news.ycombinator.com/...

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AI is dramatically accelerating AI research

👤 yurivish 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

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Sodium-ion batteries approach cost parity with LFP

👤 doener 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.pv-magazine.com/2026/11/06/sodium-ion-batteries-approach-cost-parity-with-lfp-with-70-gwh-covered-by-supply-agreements/ <p...

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Libghostty 1.0

👤 mfiguiere 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

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RealPage rent-pricing case wraps with nearly $360M to renters

👤 raybb 📅 2026-11-06 📖 1 min 📝 13 words HACKER NEWS

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I Toured Unitree's Factory. I'm More Bullish on the West

👤 josefchen 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

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The AI sell-off compared to well-known stock market crashes

👤 hubraumhugo 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.kadoa.com/companies/insights/ai-selloff-perspective Comments URL: https://news.ycombinator.com/item?id=50401029 Points: ...

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Lighting the Skies

👤 dverlaeckt80 📅 2026-11-07 📖 1 min 📝 13 words HACKER NEWS

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Plasma p-tau181, Aβ1-42, and GFAP across Alzheimer's disease, mixed-type Alzheimer's disease, mild cognitive impairment, anxiety disorders, and depressive disorders

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Haifeng Lou, Chenyun Huang, Ji Shen, Dake Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004683?dgcid=rss_sd_all

RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

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Reduced hypothalamic [³H]PK11195 binding in Lewis rats after chronic high-intensity interval training

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Atiqur Hasany, John R. Tyler, Huy Lu, Kenneth Blum, Wei Sun, Richard J. Salvi, Igor Elman, Panayotis K. Thanos

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https://www.sciencedirect.com/science/article/pii/S0006899326004622?dgcid=rss_sd_all

Improved motor learning in adenosine A1 receptor knockout mice

📖 1 min 📝 19 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maxx Freed, Noah Raffone, Devin Wilke, R. Holly Fitch, Maxim Volgushev

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Feasibility of an 8-week lifestyle intervention combining exercise, diet, cognitive training, and social engagement in individuals with mild cognitive impairment: a pilot randomized controlled trial

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Cai-Lian Wang, Hong Chen, Ruo-Yu Xu, Na Wei, Xiao-Qin Su, Jing Li, Ci Li, Jun-E Liu, Yu-Chen Qiao

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Hesperidin from Citrus reticulata peel improves welfare in PTU-induced hypothyroid rats through home-cage behavior, endocrine, neurobehavioral, and anti-inflammatory mechanisms

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Bassant A. Elbaz, Kholoud A. Ali, Samar M. Mouneir, Marwa A. Ibrahim, Essam Abdel-Sattar, Riham A. El–Shiekh, Manal A. Fouad

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Plasma miR-21-5p, miR-195-5p, and clusterin profiles in relapsing-remitting multiple sclerosis and their relationship with cognitive function

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Osman Oğuzhan Kılıç, Semra Mungan, Birsen Can Demirdöğen

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Obesity-induced oxidative stress and cGMP depletion suppress cerebellar α2,3-Na⁺,K⁺-ATPase activity in female mice

📖 1 min 📝 61 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Martina Raissa Ribeiro, Geovanni de Morais Lima, Amanda Siena, Jacqueline Alves Leite, Victoria Francelino Alves, Vitoria Cristina Araujo, André Schmidt Suaiden, Bruno de Souza Gonçalves, Marina Marq...

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Clip-to-trial: EEG-based decoding network with long short-term self-attention for DRDS recognition

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Lili Shen, Yiming Wu, Zihao Yang, Xinyuan Di

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Intranasal insulin improves long-term cognitive deficits and protects hippocampal damage following neonatal hypoxia–ischemia in juvenile rats

📖 1 min 📝 41 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Nilesh Dankhara, Jonathan W. Lee, Norma B. Ojeda, Michelle A. Tucci, Silu Lu, Elizabeth L. White, Lindsey E. Winborne, Sharon Xu, Carolyn K. Glendye, Lu-Tai Tien, Yi Pang, Abhay J. Bhatt, Lir-Wan Fan</...

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Characterization of the hemodynamic response function in default mode network subregions for task-evoked negative BOLD responses

📖 1 min 📝 29 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Hengda He, Bardiya G. Yazdi, Amirreza Sedaghat, Seyed Hani Hojjati, Sindy Ozoria, Peter Chernek, Jenseric Calimag, Farnia Feiz, Qolamreza R. Razlighi

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A multimodal MRI study of DTI-ALPS and cerebral blood flow changes across the Alzheimer's disease continuum

📖 1 min 📝 38 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yian Gao, Manxi Xu, Peiling Ou, Haoyu Yang, Jia Wang, Wulong Liu, Yu Sun, Wenying Du, Yuli Wang, Tongxi Liu, Hui Xia, Liping Fu, Ni Shu, Zhen Yuan, Guolin Ma

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Cross-excitation in dorsal root ganglia redistributes coding capacity from stimulus intensity to stimulus location

👤 Perevozniuk, D., Koshev, N., Musienko, P. 📅 2026-11-06 📖 1 min 📝 171 words

BIORXIV NEUROSCIENCE

> **Summary:** The phenomenon of cross-excitation between neurons of dorsal root ganglia (DRGs) of mammals is widely known, yet its exact mechanism remains underexplored. In this work, we analyze the phenomenon as part of the signal processing performed by the body's somatosensory system. We argue that, if the mec...

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Towards an analytical framework for investigating the neurophysiological influence of social context in speech perception

Ip, E. Y. J., Igoe, A., Cowan, B. R., Di Liberto, G. M.

2026-11-06 1 min 244 words

BIORXIV NEUROSCIENCE

Summary: Social information plays an important role in everyday speech communication, yet existing neurophysiological approaches provide limited means of isolating its contribution during naturalistic speech perception. Here, we present an analytical framework that combines the large language model (LLM) Mis...

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Extended Pavlovian conditioned approach training and devaluation in Long-Evans rats using time-restricted feeding

Rainer, I. L., Brooks, K., Ulrich-Lai, Y. M., McReynolds, J. R.

2026-11-06 1 min 246 words

BIORXIV NEUROSCIENCE

Summary: Pavlovian conditioned approach (PCA) is a useful tool in studying compulsive- and addictive-like behaviors. PCA pairs a cue with reward delivery and rats typically differentiate into three phenotypes (sign-trackers, goal-trackers, and intermediates) based on behavior during the reward-paired cue pre...

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Walking delays auditory processing in humans alongside a global shift in aperiodic neural activity

Chen, X., Chen, S., Zhang, X., Cao, L.

2026-11-06 1 min 153 words

BIORXIV NEUROSCIENCE

Summary: Natural sensory processing unfolds during movement, yet human electrophysiology is usually measured under enforced stillness. Here, we used mobile EEG to test whether walking changes the timing of auditory processing. Thirty adults listened to lateralised pure tones while walking freely or standing....

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Neonatal NMDA receptor blockade disrupts prefrontal cortical regulation of social affective behavior: implications for schizophrenia

Tachihara, R., Hayashi, N., Rosal, C. G. d., Xiong, T., Igarashi, K., Ozawa, T., Hori, M., Yamada, K., Toyoshima, M.

2026-11-06 1 min 214 words

BIORXIV NEUROSCIENCE

Summary: Developmental N-methyl-D-aspartate receptor (NMDAr) hypofunction has been implicated in the pathophysiology of schizophrenia, including cognitive and emotional deficits. However, the neural circuit mechanisms linking developmental NMDAr hypofunction to social affective impairments remain poorly unde...

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A Framework for SPICE MRSI Reconstruction Posterior Uncertainty Quantification

Lyu, T., Jbabdi, S., Finney, S. M., Clarke, W. T.

2026-11-06 1 min 166 words

BIORXIV NEUROSCIENCE

Summary: Purpose: Low-rank reconstruction methods such as SPICE substantially improve Magnetic Resonance Spectroscopic Imaging (MRSI) reconstruction quality. However, SPICE does not provide an estimate of reconstruction uncertainty. A framework for posterior uncertainty estimation in SPICE reconstruction is ...

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Curved geometry shapes the observability of hippocampal oscillations

Makarova, J., Munoz-Arnaiz, R., Ashouri, F., Herreras, O.

2026-11-06 1 min 162 words

BIORXIV NEUROSCIENCE

Summary: Curved neural architectures reshape the spatial distribution of field oscillations, complicating the identification of their underlying generators. To test these principles experimentally, here we resolve the paradoxical poor expression of theta oscillations in the hilus of the folded dentate gyrus ...

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A biological light sheet links cellular voltage to mesoscopic brain dynamics

Chen, T.-W., Lin, B.-J. 2026-11-06 1 min 135 words BIORXIV NEUROSCIENCE

Summary: Membrane voltage is the fundamental currency of neural computation, yet cellular-resolution voltage imaging of large neuronal populations in vivo remains limited by trade-offs among imaging speed, field of view, and optical sectioning. We overcome this challenge by encoding a biological light sheet ...

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Persistent hippocampal traces of lifelong spatial memory

Wu, J., Meister, M., Lois, C. 2026-11-06 1 min 234 words BIORXIV NEUROSCIENCE

Summary: Memory presents a biological paradox: the brain changes continuously, yet a brief experience can influence behavior for a lifetime. How such memories are stored in brain circuits remains unclear. Here we show that two days in a binary-tree maze leave mice with spatial memories that survive repeated ...

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Next generation Human Neocortical Neurosolver model with biophysically realistic somatic, dendritic and synaptic dynamics

Duecker, K., Soplata, A., Tolley, N., Pugliese, S., Jones, T. A. E., Kandpal, C., Waller, D. A., Daniels, D. S., Jones, S. R.

2026-11-06 1 min 221 words BIORXIV NEUROSCIENCE

Summary: The Human Neocortical Neurosolver (HNN) is an open-source biophysical modeling tool for multiscale interpretation of human electroencephalography (EEG) and magnetoencephalography (MEG) signals. HNN's foundation is a pre-tuned canonical neocortical circuit model consisting of excitatory pyramidal neu...

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Phase aligned EEG source space features for distinguishing Alzheimer’s disease from frontotemporal dementia

2026-11-06 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-79695-5

A topographic video model predicts the spatio-functional organization of the human visual system

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https://www.nature.com/articles/s41467-026-78948-1

The subcommissural organ secretes SCO-spondin, associated with stress response and mood

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https://www.nature.com/articles/s42003-026-11220-w

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

Yingjie Zhu 2026-11-03 1 min 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

Ole A. Andreassen 2026-11-06 1 min 25 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 06 November 2026; doi:10.1038/s41593-026-02536-1 Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02536-1

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

👤 Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

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https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Neuroimmune-endocrine interactions in perinatal asphyxia: from brain injury to neuroprotection

Francisco Capani 2026-11-06 1 min 295 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perinatal asphyxia (PA) remains a major cause of neonatal mortality and long-term neurodevelopmental disability worldwide. Although hypoxic–ischemic brain injury has traditionally been attributed to oxygen deprivation, excitotoxicity, and oxidative stress, accumulating evidence indicates that neurol...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2018788

Peripheral physiological decoding of arousal in a boundary-avoidance task: signal validity and individualized control

Anirudh Natarajan and Paul Sajda 2026-11-03 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Scoring well on a task-demand label does not show that a decoder measures arousal. We ask whether peripheral physiology decodes demand better than EEG, whether the index behaves like graded arousal rather than a clock, and what a controller could do with it. Approach. We re-analysed a pub...

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GROND: automated characterization of periodic and rhythmic EEG patterns

Jin Jing, Chenxi Sun, Tianyu Zhang, Matthew Byrd, Alexandra–Maria Tăuţan, Lara Basovic, Peter N Hadar, Marta P Fernandes, Daniel Goldenholz, Jennifer Kim, Aaron F Struck, Sahar F Zafar and M Brandon Westover

2026-11-03 2 min 469 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Periodic discharges (PDs) and rhythmic delta activity (RDA) are common electroencephalographic patterns in critically ill patients that require detailed characterization—including lateralization, spatial localization, and frequency estimation—according to the American Clinical Neurophysio...

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Transformer networks are robust learners of source localization for EEG measurements

David Hart, Simon Polishchuk and Moritz Dannhauer

2026-11-06 1 min 356 words JOURNAL NEURAL ENGINEERING

Summary: Objective. EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade un...

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ACF-Net: design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang
2026-11-06 1 min 292 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Cross-subject gesture recognition in wearable brain–computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, a...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb723

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Alon Gorenshtein, Yosef Adiniaev, Tom Liba, Yiftach Barash, Eyal Klang and Oved Daniel

2026-11-06 1 min 297 words JOURNAL NEURAL ENGINEERING

Summary: Objective. A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts. Approach. Retrospective secondary analysis of BigP3BCI 1...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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First-in-Human Regulatory T Cell-Derived Exosomes Plus Ranibizumab for Refractory nAMD

Yong Tao 2026-11-05 1 min 59 words LOW VISION

Summary: Persistent fluid despite intravitreal anti-vascular endothelial growth factor (VEGF) treatment remains common in neovascular age-related macular degeneration (nAMD). Regulatory T cell-derived exosomes (TregExo) have preclinical immunomodulatory activity. To evaluate the safety and maximum tolerated ...

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Bias-reconfigurable barrier photodetectors for integrated broadband sensing and neuromorphic vision

Hongwei Liu 2026-11-05 1 min 65 words LOW VISION

Summary: Intelligent vision systems demand photodetectors with reconfigurable response to support diverse modalities like high-speed sensing and neuromorphic processing. Here we show a bias-reconfigurable barrier photodetector that enables continuous modulation of response time across 7 orders of magnitude, ...

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Clinical Context-Dependent Value of bpMRI-Derived Imaging Representations for Prostate Cancer Risk Stratification

👤 Xiaojing He 📅 2026-11-05 📖 1 min 📝 60 words LOW VISION

Summary: The objective was to evaluate the clinical context-dependent value of biparametric MRI (bpMRI)-derived imaging representations for prostate cancer (PCa) risk stratification, with emphasis on whether imaging features provide complementary information in patients within the prostate-specific antigen (...

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Cross-Center Stroke Evaluation using Carotid Ultrasound Patterns via Scale-Adaptive Frequency-Enhanced Neural Network

👤 Peng-Hsiang Hung 📅 2026-11-05 📖 1 min 📝 61 words LOW VISION

Summary: Medical imaging datasets vary across institutions due to differences in scanner types, posing challenges for model generalization. To address the issue, this study proposed a scale-adaptive frequency-enhanced neural network (SAFE-Net), which leveraged a DWT-ViT encoder to incorporate frequency-domai...

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Screen-detected health conditions among students in grade 1-9: findings from Timor-Leste's first nationwide school-based screening program

👤 Mokryeon Cho
📅 2026-11-05 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: Dental caries and low BMI-for-age were the most frequently screen-detected problems. For the dental assessments the screening team accounted for more of the variation than the municipality, and this variation was specific to the dental tasks rather than to the examiners, who were consis...

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A novel auxiliary method combining a smartphone with a handheld slit-lamp for ocular image documentation of cataracts and diabetic retinopathy in remote regions

👤 Qibin Xu 📅 2026-11-05 📖 1 min 📝 53 words LOW VISION

Summary: CONCLUSIONS: All these results demonstrate the preliminary technical feasibility of this method, which may serve as a practical auxiliary tool for preliminary ocular image acquisition and documentation in remote and low-resource settings, enabling ophthalmic review of disease, teleconsultation by op...

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Exploring cerebral visual impairment: functional vision classification and caregiver concerns

👤 Manish Poudel 📅 2026-11-05 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSION: Caregiver-reported concerns differed descriptively across age groups and functional vision phases. Integrating caregiver observations with functional vision assessment may provide a more comprehensive understanding of visual difficulties and support individualized rehabilitation planning...

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Co-evaluating health messages through inclusive participatory judging and validation: A public involvement study in Tanzania's eye health education programme

Ving Fai Chan 2026-11-05 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Participatory judging and validation offered an inclusive and empowering approach to public involvement in health communication design. Beyond improving message quality, these processes fostered dialogue, shared decision-making and community ownership - demonstrating a transferable mode...

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Assessment of visual and cognitive function among patients with Stargardt Disease

Sharqa Fatima 2026-11-06 1 min 72 words LOW VISION

Summary: This study aimed to assess the association between visual and cognitive impairment in patients with Stargardt Disease at the Low Vision Rehabilitation Centre, Al-Shifa Trust Eye Hospital, Rawaalpindi. It was a cross-sectional study conducted from September to December 2023 which included 32 patients...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

👤 AABC Study Group
📅 2026-11-06 📖 1 min 📝 74 words

LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The interplay of neural networks enabling social interaction: Protocol of a multi-level, mixed-methods study of empathy and theory of mind

Lara Z Maliske
2026-11-05 1 min 73 words FNIRS

Summary: BACKGROUND: The quality and frequency of social interactions are among the best predictors of long-term mental and physical health. So how do good social interactions arise? Social neuroscience has focused on a number of affective and cognitive functions including empathy and Theory of Mind (ToM) as...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Augmented reality paradigm and EA-DGCNN decoding algorithm for motor imagery in stroke patients

Tzyy-Ping Jung 2026-11-04 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Stroke represents one of the leading causes of long-term motor dysfunction on a global scale. Existing motor imagery (MI) brain-computer interface (BCI) rehabilitation training systems face critical challenges of suboptimal patient engagement and limited electroencephalography (EEG) featu...

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Perception gaps between patients and practitioners regarding safety, data security, and ethics of intracranial electrode implantation: a multi-center study

👤 Huanguang Liu
📅 2026-11-04 📖 1 min 📝 44 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: DBS patients systematically underestimate risks compared to practitioners, driven partly by knowledge deficits that current informed consent processes do not address. These findings provide empirical evidence for redesigning consent practices and developing governance frameworks for DBS...

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A human rheumatoid arthritis bone-cartilage interface organoid recapitulates bone, cartilage, and immune interactions

👤 Jessica Cottrell 📅 2026-11-05 📖 1 min 📝 55 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This modular, human-derived RA BCI organoid reproduced key features of RA pathogenesis that can enable direct assessment of therapeutic mechanisms at the bone-cartilage-synovial interface. The RA BCI organoid provides a physiologically relevant, scalable platform for preclinical drug di...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

Ximo Wang 2026-11-05 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

Johnsi Inbakumari 2026-11-05 1 min 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

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👤 /u/ scartfife 📅 2026-11-06 📖 1 min 📝 60 words REDDIT PYTHON

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Anthropic IPO valuation talks drop to $1.3T

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Kalshi court losses pile up as judges refuse to block state gambling laws

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Memorilla: Latent Semantic Memory for LLMs

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We Know 15,579 Websites Are Infected. How Do We Tell Them?

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Category differences in similarity between fixation density maps and psychological potential fields for sketches

👤 Naoyuki Awano, Yuki Hayashi 📅 2026-11-06 📖 1 min 📝 249 words ARXIV CS HC

Summary: arXiv:2611.06930v1 Announce Type: cross Abstract: When an object comes into view, humans can instantaneously recognize it if known to them. However, assessing how this recognition takes place remains challenging. A few studies using simple sketches have reported varying degrees of image similarity ...

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https://arxiv.org/abs/2611.06930

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Souffleur: A Glanceable Display for Mobile Voice Conversations

Anton Wolter, William Benjamin, Niklas Elmqvist

2026-11-06 1 min 124 words

ARXIV CS HC

Summary: arXiv:2611.08443v1 Announce Type: new Abstract: The speech capabilities of mobile devices have grown, but their displays still rely on a text-based chat. When resuming a voice chat after a context switch, the last messages may not convey the key points of the conversation. Autobiographical use and ...

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https://arxiv.org/abs/2611.08443

ControlAnything: Generative UI and On-Demand Device Integration for Mixed-Reality Hand Interaction by a Coding Agent

Bektur Ryskeldiev

2026-11-06 1 min 252 words

ARXIV CS HC

Summary: arXiv:2611.08321v1 Announce Type: new Abstract: Connecting hand input in mixed reality (MR) to a nearby physical device typically requires an integration written in advance for that device, its transport and the mapping. ControlAnything defers this integration to the moment of use and assigns it to...

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IntentWeaver: Sustaining Coherence across Evolving Intents in Human-AI Co-Writing

Yuanhao Zhang, Xinru Cheng, Baiqiao Zhang, Xinqi Zhang, Assem Zhunis, Shuai Ma, Xiaojuan Ma

2026-11-06 1 min 154 words ARXIV CS HC

Summary: arXiv:2611.08214v1 Announce Type: new Abstract: Writers hold intents at many levels, from overarching goals to concrete wording, and evolve them by periodically popping back up from the text to higher-level intents. Generative AI co-writing disrupts this practice. Dense drafts capture attention and...

Read full article:
https://arxiv.org/abs/2611.08214

Reflective Nature Experiences in Virtual Reality

Shiva Ghasemi, Maddie Oehler, Annie Xenakis, Jeffrey Jiarui Yang, Scott McCrickard

2026-11-06

1 min 122 words ARXIV CS HC

Summary: arXiv:2611.08167v1 Announce Type: new Abstract: This paper presents the design and evaluation of an accessible virtual reality (VR) nature experience developed to promote mindfulness, reduce stress, and support emotional well-being. The system immerses users in a calming natural environment through...

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Tasteply: Externalizing Stable and Situational Preference as Manipulable Spatial Objects

DaEun Choi, Daniel Lee

2026-11-06 · 1 min · 137 words · ARXIV CS HC

Summary: arXiv:2611.07851v1 Announce Type: new Abstract: Most recommender systems treat user preferences as a single signal to infer and optimize, yet human preferences are a mix of a stable preference and situational preferences that turn on and off depending on context. Through Tasteply, we externalize th...

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https://arxiv.org/abs/2611.07851

VocabulARy 2.0: A Generative AR System Enabling Personalised 3D Mnemonics for Vocabulary Learning in Context

Karolina Trajkovska, Matja\v{z} Kljun, Maheshya Weerasinghe, Klen \v{C}opi\v{c} Pucihar

2026-11-06 · 1 min · 108 words · ARXIV CS HC

Summary: arXiv:2611.07798v1 Announce Type: new Abstract: Mnemonics support vocabulary learning by linking unfamiliar words to familiar concepts through memorable verbal and visual associations. We present a generative AR system that enables learners to create personalised 3D mnemonics by transforming famili...

🔗 **Read full article:**
https://arxiv.org/abs/2611.07798

Thinking Is Not Writing: Designing AI Chatbots to Connect Reasoning and Revision

Lucile Favero, Tanja K\"aser, Juan Antonio P\'erez-Ortiz, Nuria Oliver

2026-11-06 1 min

157 words

ARXIV CS HC

Summary: arXiv:2611.07792v1 Announce Type: new Abstract: Supporting students' argumentative writing with AI requires connecting reflection to their revision processes. We present Maike, a Socratic chatbot supporting writing through argument diagnosis, targeted questioning, and a student model across Think a...

🔗 **Read full article:**
https://arxiv.org/abs/2611.07792

Breaking the Modality Barrier: Multimodal Fusion With Missing Modalities for Cybersickness Prediction

Purnata Saha, Rifatul Islam, John Quarles

2026-11-06 1 min 154 words

ARXIV CS HC

Summary: arXiv:2611.07386v1 Announce Type: new Abstract: Cybersickness is a major barrier to comfortable VR use, and reliable prediction often requires multiple physiological and behavioral sensors. However, existing models typically assume a fixed set of modalities, making them vulnerable when sensors are ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.07386

Slide to Report: Continuous Reporting Symptom-Specific Cybersickness in VR

Purnata Saha, Rifatul Islam, Qiliang He, John Quarles

2026-11-06 1 min 156 words

ARXIV CS HC

Summary: arXiv:2611.07345v1 Announce Type: new Abstract: Cybersickness is often assessed during VR exposure through discrete overall sickness, which cannot capture specific symptoms or their changes over time. Post-exposure questionnaires such as the Simulator Sickness Questionnaire provide symptom-specific...

🔗 **Read full article:**
https://arxiv.org/abs/2611.07345

BrainPFN: Amortized Brain-Behaviour Prediction via Prior-Data Fitted Networks

Carlo Alberto Barbano (MIND), Demian Wassermann (MIND)

2026-11-06 1 min 244 words

ARXIV QBIO NC

Summary: arXiv:2611.06934v1 Announce Type: new Abstract: The standard approach to predicting behavioral phenotypes from functional MRI is represented by Kernel Ridge Regression (KRR) applied on functional connectivity (FC) matrices. However, this approach is limited: separate models must be fit and tuned fo...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06934

Capturing Membrane Dynamics and Spike Timing of Human Neurons at Scale using Neural Operators

Luca Ghafourpour, Cindy Zhao, Valentin Duruisseaux, Bahareh Tolooshams, Philip H. Wong, Costas A. Anastassiou, Anima Anandkumar

2026-11-06 1 min 211 words ARXIV QBIO NC

Summary: arXiv:2611.06933v1 Announce Type: new Abstract: Characterizing the properties of neuronal cell types is central to understanding brain function. Yet, biophysically realistic neuron models remain difficult to scale to large populations while preserving the richness of cellular dynamics and within-ce...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06933

Oscillatory Phase Coding Emerges in Adaptive RNNs for Flexible Working Memory

Kehan Lu, Tianhao Chu, Aiersi Tuerhong, Yiwen Sun, Huan Luo, Fang Fang, Si Wu

2026-11-06 1 min 215 words ARXIV QBIO NC

Summary: arXiv:2611.06932v1 Announce Type: new Abstract: Oscillatory phase coding (OPC) is widely observed in neural systems and has been implicated in cognitive functions such as memory and sequence processing. However, its computational role and neural substrates remain largely unclear. Here, we studied O...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06932

Distilling Neural Population Geometry into Transformers and GRUs for Motor Control

Maryna Kapitonova, Tonio Ball
2026-11-06 1 min 197 words ARXIV QBIO NC

Summary: arXiv:2611.06931v1 Announce Type: new Abstract: We introduce neuro-distillation (Neuro-Dist), a technique designed to directly transfer computationally useful neural population structure from biological brains into artificial controllers. Using three macaque motor-cortical datasets as neuro-distill...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06931

ApproxBrain: Certified Sufficiency Frontiers in Connectome-Constrained Neural Systems

Ali Mahdavi, Azadeh Zamanifar, Amirfarhad Farhadi, Omid Kashefi
2026-11-06 1 min 268 words ARXIV QBIO NC

Summary: arXiv:2611.06929v1 Announce Type: new Abstract: Background. A connectome specifies the substrate available to a nervous system, not how much of it a given computation requires. Reported compact sufficient circuits are point estimates: one subset, one task, one fidelity threshold, one parameterisati...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06929

A minimal dynamical model of human-AI cognitive coevolution

Petar Jolakoski, Dimitar Trajanov, Qiyu Sun, Zhen Wu, Ljupco Kocarev

2026-11-06 1 min

244 words

ARXIV QBIO NC

Summary: arXiv:2611.06928v1 Announce Type: new Abstract: The rapid deployment of generative AI raises the question of whether pervasive cognitive offloading drives a measurable, potentially irreversible decline in unassisted human reasoning. We examine this question with a phenomenological dynamical model b...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06928

Reading a neuron's tuning from its correlations with other neurons

Francesco L\"assig

2026-11-06 1 min 303 words

ARXIV QBIO NC

Summary: arXiv:2611.06927v1 Announce Type: new Abstract: What a neuron represents is usually established by relating its activity to a stimulus. We ask whether the same facts are contained in the correlation matrix of a recorded population, which keeps the pairwise relations among the neurons and discards t...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06927

Foundation Models as Modular Tools for fMRI Analysis

Joseph Kilgore, Noemia Mahmud, Shinjae Yoo, David Keetae Park

2026-11-06 1 min

259 words

ARXIV QBIO NC

Summary: arXiv:2611.06926v1 Announce Type: new Abstract: Foundation models (FMs) for the brain are rapidly growing in number and each captures a different aspect of the same data. A single scientific question often needs several of them at once, alongside classical analysis and the literature, but no establ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06926

No Support Beyond Reach: The Measure Centrality Index 2.0 and Adversarial Comparisons of Consciousness Theories

Robert Chis-Ciure

2026-11-06 1 min 249 words

ARXIV QBIO NC

Summary: arXiv:2611.06925v1 Announce Type: new Abstract: A success in adversarial testing of scientific theories reaches only as deep as its matching failure would have cut. I call this the Reach Principle, and the goal of this chapter is to build the instrument that enforces it. Adversarial tests of consci...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06925

CAG repeat length shapes bidirectional functional connectome alterations in spinocerebellar ataxia type 3: a multimodal MRI study

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Xianfei Yang, Haishuo Xia, Yuwei Hu, Can Xu, Xingang Wang, Qingxin Deng, Qian Xie, Limeng Dai, Qifang Yang, Changying Li, Li Gui, Chen Liu

🔗 **Read full article:**
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Coarse motor-state structure across paired preoperative fMRI and intraoperative ECoG in patients with brain tumors

📖 1 min 📝 36 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jingxi Hu, Jie Ma, Wanwan Guo, Yang Jiao, Jiaqi Yang, Hongwei Wang, Yiheng Yin, Renke He, Gang Cheng, Hongmin Bai, Hairong Zheng, Dong Liang, Jianning Zhang, Zhanli Hu

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https://www.sciencedirect.com/science/article/pii/S1053811926006257?dgcid=rss_sd_all

Learning dynamic affinity diffusion for multi-contrast MRI super-resolution via implicit neural representations

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Huajie Wen, Lixin Liang, Rashid Khan, Inayatul Haq, Kai Chen, Lin Lin, YongSheng Liang, Bingding Huang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006397?dgcid=rss_sd_all

Use of Integrative Proteomic Analysis to Uncover Shared Molecular Signatures Across Distinct Forms of Hydrocephalus

👤 Mahendran, G., Kirupaharan, P., Audu, D., Blazer-Yost, B. L. 📅 2026-11-05 📖 1 min 📝 250 words

BIORXIV NEUROSCIENCE

Summary: Hydrocephalus is a heterogeneous disease characterized by disruption of cerebrospinal fluid (CSF) dynamics and can arise from diverse etiologies, including post-hemorrhagic hydrocephalus (PHH), hydrocephalus following subarachnoid hemorrhage (SAH), post-infectious hydrocephalus (PIH), and idiopathic...

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Auditory Evoked Potentials Differ from Controls and Correlate with Aspects of Cognition in Kabuki Syndrome and Rubinstein-Taybi Syndrome

👤 Kalinousky, A., Bucklin, E., Ng, R., Meier, L., Jiddou, H., Fanara, P., Ewen, J. B., Harris, J.

📅 2026-11-06 📖 1 min 📝 268 words BIORXIV NEUROSCIENCE

Summary: Objective Kabuki syndrome (KS) and Rubinstein-Taybi syndrome (RSTS) are two disorders caused by dysfunctional components of the epigenetic machinery and both result in intellectual disability. Mouse models of both KS and RSTS have seen reversal of cognitive impairment with epigenetic treatments. How...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764287v1?rss=1

Modeling circadian rhythm splitting under chronic light and the asymmetric transition kinetics between lighting conditions

👤 Goltsev, A. V., Mendes, J. F.

📅 2026-11-06 📖 1 min 📝 231 words BIORXIV NEUROSCIENCE

Summary: In various vertebrates, chronic light splits the single daily 24-h circadian activity bout into two 12-h behavioral components per cycle, yet the underlying mechanism remains unresolved. Here, we develop a two-lobe biophysical model of the suprachiasmatic nucleus (SCN) to understand this phenomenon....

🔗 **Read full article:**
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Beyond Terrestrial Carnivores: Olfactory Bulb Anatomy in a Semi-Aquatic Mammal

Fraga-Abelleira, S. A., Torres, M. V., Canales, R., Ruiz-Rubio, S., Sanchez-Quinteiro, P., Ortiz-Leal, I.

2026-11-06 1 min 246 words BIORXIV NEUROSCIENCE

Summary: The organization of the olfactory systems remains poorly understood in mustelids, particularly in lutrines adapted to semiaquatic environments. This study provides the first detailed anatomical, histological, lectin-histochemical, and immunohistochemical characterization of the main (MOB) and access...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764355v1?rss=1

Distinct lifespan trajectories of cortical intrinsic timescales and neural oscillations

Liu, X., Harris, B., Lu, R., Wang, S., Yu, Q., Zhang, M., Cui, R., Liang, X., Fan, T.

2026-11-06 1 min 189 words BIORXIV NEUROSCIENCE

Summary: The brain integrates information over time through the persistence of neural activity, yet how this persistence changes across the human lifespan and the mechanisms underlying these changes remain unclear. Using resting-state magnetoencephalography from three independent cohorts (n = 1,156; ages 6-8...

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https://www.biorxiv.org/content/10.64898/2026.11.04.764097v1?rss=1

Single-cell multi-omics identify OTP and NOVA1 as regulators of leptin action in hypothalamic feeding circuits

👤 Tan, H. L., Tan, Y., Ruiz-Ortiz, J., Paul, M., Han, Z., Kelly, L., Toci, A., Plucinska, K., Tajima, Y., Shiao, R., Lee, B., Ilanges, A., Darnell, R. B., Cohen, P., Friedman, J. M.

📅 2026-11-06 📖 1 min 📝 280 words BIORXIV NEUROSCIENCE

Summary: Leptin acts directly and indirectly on multiple neuronal populations to regulate feeding and numerous other physiologic functions. However, our understanding of how a single circulating signal coordinates diverse neuronal populations across multiple circuits is limited. To further elucidate how lept...

🔗 **Read full article:**
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Early AAV9-mediated SLC9A6 gene replacement preserves cerebellar structure and motor function in the shaker rat model of Christianson syndrome

👤 Dahiya, K., Sekaran, K., Hu, L., Bloomfield, A., Kapasakis, E., Henning-Smith, S., Riley, L. G., Evans, M., Haslam, S., Katakowski, J., Anderson, D. N., Gold, W., Anderson, C. J.

📅 2026-11-06 📖 1 min 📝 200 words BIORXIV NEUROSCIENCE

Summary: Christianson syndrome (CS) is an X-linked neurodevelopmental and neurodegenerative disorder caused by SLC9A6 loss of function, resulting in NHE6 deficiency and epilepsy, intellectual disability, cerebellar atrophy, and ataxia. Following on our prior gene replacement therapeutic development, we precl...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764455v1?rss=1

Personalized Prediction of Differences in fMRI Connectivity Estimates Across Preprocessing Pipelines

👤 Maximov, Y., Volkov, A., Chattopadhyay, T., Thompson, P. M., Gabitov, I.

📅 2026-11-06 📖 1 min

📝 221 words

BIORXIV NEUROSCIENCE

Summary: Longitudinal functional MRI of the human brain is valuable for tracking changes in functional connectivity (FC) associated with disease progression or treatment response. Even so, measured FC values depend on preprocessing, and nuisance regression and temporal filtering affect individuals differentl...

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https://www.biorxiv.org/content/10.64898/2026.11.04.763874v1?rss=1

Temporal Structure in Resting-State fMRI under Marginal and Cross-Spectral Surrogate Controls

👤 Volkov, A., Maximov, Y., Chattopadhyay, T., Thompson, P. M., Gabitov, I.

📅 2026-11-06 📖 1 min

📝 269 words

BIORXIV NEUROSCIENCE

Summary: Resting-state functional MRI is commonly analyzed using correlations between regional signals and their frequency spectra, which characterize average connectivity and spectral properties but may miss other aspects of temporal organization. Measures of temporal complexity offer complementary informat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.763876v1?rss=1

An expansive internal chemosensory system in the insect abdomen implicates Gustatory Receptors in diverse aspects of interoception

Fujii, S., Amrein, H.
2026-11-06 1 min 223 words BIORXIV NEUROSCIENCE

Summary: Taste and olfactory receptors are gatekeepers that convey chemical information about the external environment. Their role in chemosensory systems has been extensively studied in several model system, which established their role in triggering discrete behavioral as well as physiological responses re...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764383v1?rss=1

Test-retest reliability of resting-state functional connectivity: a methodological investigation with implications for aggression research

2026-11-06 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77588-1

Effects of chronic oxytocin administration on white matter microstructure in autistic adults: a pilot pharmaco-neuroimaging DTI study

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https://www.nature.com/articles/s41398-026-04542-0

Novel mechanism and genetic association of graft-induced dyskinesia in CRISPR rs6265 BDNF polymorphism rats

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https://www.nature.com/articles/s41531-026-01605-3

The hidden impact of cage position on behavioral phenotyping and reproducibility

📅 2026-11-06 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-79912-1

Motor imagery EEG signal classification using SAGAN-based data augmentation and ResNet-18 for brain–computer interface applications

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https://www.nature.com/articles/s41598-026-78683-z

Sympathetic neurons remodel the extracellular matrix for thermogenesis

📅 2026-11-06 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42255-026-01659-w

Triiodothyronine modulates LAT2 vesicle dynamics and potentiates β-adrenergic cAMP responses in astrocytes

📅 2026-11-06 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41420-026-03462-8

NAD+ precursor fails in Parkinson’s disease

📅 2026-11-06 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s43587-026-01277-x

Orexin and pupillary modulation for hunger control

👤 Seher KosarEva BraceyNikola GrujicYihui DuDaria Peleg-RaibsteinDenis Burdakovahttps://ror.org/02q2nz798ETH Zürich, Department of Health Sciences and Technology, Neuroscience Center Zürich, Institute for Neuroscience, Schwerzenbach 8603, Switzerland

📅 2026-10-28 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThis study reveals a genetically defined arousal-hunger circuit that enables mental engagement or external sensory input to naturally interrupt the urge to eat. These findings in mice suggest that...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619759123?af=R

Dlx1&2 directly promote expression of Cxcl14 to control synapse development and interneuron survival

👤 Teng GuoSusan LindtnerJia Sheng HuSiavash Fazel DarbandiYurong WangEmily Ling-Lin PaiAthéna R. YpsilantiCindy T. J. van VelthovenAlex S. NordJohn L. R. RubensteinaNina Ireland Laboratory of Developmental Neurobiology, Department of Psychiatry and Behavioral Sciences, University of California San Francisco, San Francisco, CA 94143bDepartment of Neurobiology, Physiology, and Behavior, Center for Neuroscience, University of California Davis, Davis, CA 95618cDepartment of Psychiatry and Behavioral Sciences, Center for Neuroscience, University of California Davis, Davis, CA 95618dhttps://ror.org/03cpe7c52Allen Institute for Brain Science, Seattle, WA 98109

📅 2026-10-28 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDlxgenes control multiple steps in the development of cortical interneurons (CIN), including synapse development and cell survival. Here we explore molecular mechanisms how theDlxgenes promote neo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2616055123?af=R

Lysine crotonylation acts as an epigenetic switch for glutamate neurotransmission and spatial memory

👤 Utsav MukherjeeBudhaditya BasuPratyush SuryavanshiStacy E. BeyerSaaman GhodsiNathan RobillardYann VanrobaeysEric B. TaylorJoseph GlykysTed AbelSnehajyoti Chatterjeeahttps://ror.org/036jqmy94Department of Neuroscience and Pharmacology, Carver College of Medicine, University of Iowa, Iowa City, IA 52242bhttps://ror.org/036jqmy94Iowa Neuroscience Institute, University of Iowa, Iowa City, IA 52242chttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Neuroscience, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Stead Family Department of Pediatrics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Genetics, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Fraternal Order of Eagles Diabetes Research Center, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Department of Molecular Physiology and Biophysics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242

📅 2026-10-28 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAlthough histone acetylation has long been considered a principal regulator of gene expression events underlying memory storage, the role of non-acetyl histone acyl modifications on long-term memo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2608460123?af=R

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Short excitatory projections promote burst activity and modular functional organization in neural networks

Claus C. Hilgetag 2026-11-03 1 min 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1965686

Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

George Giakos 2026-11-03 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.2030505

Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

Makoto Nakane 2026-11-03 1 min 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1940507

Persistent memory through triple-loop consolidation under stochastic unit turnover

👤 Jianwei Lou 📅 2026-11-03 📖 1 min 📝 366 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1845213

Editorial: Advances in perceptual decision making and brain oscillations

👤 Francesco Di Gregorio 📅 2026-11-03 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Temporal dissociation across the eye-brain-muscle cascade delineates a preparation window before muscle activation in subacute stroke

Jing Tao 2026-11-06 1 min 238 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Upper-limb motor impairment after stroke is compounded by a disrupted sensorimotor cascade linking visual attention, cortical motor preparation, and peripheral muscle activation. Whether these stages are temporally dissociated in subacute stroke, and whether their timing carries information about th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2002490

Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1960197

From beat to syntax: evolutionary and developmental paths of a shared neural resource for hierarchical processing in music and language

Tao Gong 2026-11-03 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Children who track a musical beat well tend to have stronger grammatical skills, and brief exposure to a regular rhythm can transiently improve grammatical processing. These findings raise the possibility of a shared underlying mechanism, and with it two questions: where did the link come from, and ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1967182

Vagus nerve stimulation in obesity: a systematic review and meta-analysis

Matthew A. Schiefer 2026-11-03 1 min 333 words FRONTIERS NEUROSCIENCE

Summary: BackgroundVagus nerve stimulation (VNS) has been investigated as a potential therapy for obesity because of the vagus nerve’s role in satiety, gastrointestinal signaling, autonomic regulation, and metabolic homeostasis. However, studies vary substantially in stimulation site, interface, parameters, ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1955378

Editorial: Expanding the therapeutic dose range of cannabinoid-based therapy

Babak Baban 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2028326

Analysis and improvement of surface electromyography-based American Sign Language hand gesture recognition

Yasuharu Koike 2026-11-03 1 min 243 words FRONTIERS NEUROSCIENCE

Summary: Surface electromyography (sEMG) allows sign language gestures to be recognized from forearm muscle activity without visual access to the hand. Gestures that produce similar forearm activation patterns, however, remain difficult to distinguish. This study examined how these similarities relate to cla...

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Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo
2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb442

Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32e

CORRIGENDUM: In silico development and validation of Bayesian methods for optimizing deep brain stimulation parameters (2023 J. Neural Eng. 20 036015)

Sumedh S Nagrale, Ali Yousefi, Theoden I Netoff and Alik S Widge

2026-10-29 1 min

0 words

JOURNAL NEURAL ENGINEERING

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32d

REFS: robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

Xueyuan Xu, Wenjia Dong, Fulin Wei and Li Zhuo

2026-11-06 1 min 268 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Contrast Sensitivity Functions in ON and OFF Visual Pathways in Myopia

👤 Jiawei Zhou 📅 2026-11-05 📖 1 min 📝 49 words LOW VISION

Summary: CONCLUSIONS: High myopia exhibited reduced ordinary and ON pathway CS. The ON pathway-selective CS deficits were significantly correlated with myopia severity and predominantly observed at low SFs. These findings provide a theoretical basis for understanding visual function in myopia, developing eff...

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Semimetallic PtSe2 Phototransistors with Back-to-Back Ohmic Contacts for Low-Power Neuromorphic Vision and Polarization-Selective Detection

👤 Pengfei Hou 📅 2026-11-05 📖 1 min 📝 56 words LOW VISION

Summary: The rapid advancement of artificial intelligence has created an urgent demand for energy-efficient neuromorphic vision, positioning phototransistors as a promising platform for next-generation artificial visual synapses. However, conventional phototransistors are limited by interfacial defects—inclu...

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Multisegment spectacles for prevention of myopia progression

Bettina Wabbels 2026-11-05 1 min 57 words LOW VISION

Summary: BACKGROUND: The prevalence of myopia is postulated to be rising worldwide, along with the risk of secondary complications, such as maculopathy; however, the evidence for this remains inconclusive, particularly for Europe. The focus is on reducing the increased axial length growth in childhood. To th...

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The epidemiology of frailty in urban and rural KwaZulu Natal, South Africa; a cross-sectional study

Bilkish Cassim 2026-11-05 1 min 29 words LOW VISION

Summary: CONCLUSION: Frailty is common and associated with poorer HRQoL. The identified clinical risk factors can be used to prioritise individuals for referral for clinical assessment of frailty in SA.

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A lightweight vision-based framework for automated aero-engine blade damage inspection

Zhengwei Zhang 2026-11-05 1 min 71 words LOW VISION

Summary: Automated visual inspection of aero-engine blades is central to aviation safety, yet it is challenged by damage that is small and low-contrast against a strongly textured metal surface, and by the need for models that are light enough for edge deployment. This paper presents a lightweight vision-bas...

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Prevalence of Burnout in Cardiothoracic Surgery From Training to Consultancy: A Meta-Analysis

Tomasz Płonek 2026-11-05 1 min 30 words LOW VISION

Summary: CONCLUSIONS: Burnout is highly prevalent across all stages of a cardiothoracic surgical career. The observed geographic variation warrants further investigation into whether systemic and cultural factors may contribute to burnout.

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Surface Glare Suppression in Retinal Imaging With a Cross-Polarized Line-Scanning Color Fundus Camera

Yaochun Shen 2026-11-05 1 min 68 words LOW VISION

Summary: Fundus imaging is critical for the early detection and management of retinal diseases, yet cost and technical limitations remain barriers to broad implementation, especially in resource-limited settings. Conventional fundus photography and scanning laser ophthalmoscopy (SLO) involve trade-offs in co...

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Multisensory interactions between nociception and vision through the looking glass

Lieve Filbrich 2026-11-05 1 min 77 words LOW VISION

Summary: To efficiently react to a potential threat to protect the body's integrity, it is not only crucial to localize nociceptive stimuli on the body, but also the position of the potentially harmful objects in external space. It is therefore essential to coordinate the representation of the body and that ...

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Combined Versus Monotherapy Anti-Complement and Anti-Vascular Endothelial Growth Factor Intravitreal Injections for Age-Related Macular Degeneration: Safety Outcomes in the Real World

👤 Edward H Wood
📅 2026-11-05 📖 1 min 📝 59 words

LOW VISION

Summary: CONCLUSION: In this analysis of outcomes from routine clinical practice, dual anti-VEGF and anti-complement therapy was not associated with higher overall rates of serious ocular adverse events compared with monotherapy. Serious ocular adverse events were rare across all treatment strategies. Additi...

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First-in-Human Regulatory T Cell-Derived Exosomes Plus Ranibizumab for Refractory nAMD

👤 Yong Tao 📅 2026-11-05 📖 1 min 📝 59 words LOW VISION

Summary: Persistent fluid despite intravitreal anti-vascular endothelial growth factor (VEGF) treatment remains common in neovascular age-related macular degeneration (nAMD). Regulatory T cell-derived exosomes (TregExo) have preclinical immunomodulatory activity. To evaluate the safety and maximum tolerated ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu 2026-11-04 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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A Neuro-endocrine-immune Network Perspective on Transcranial Direct Current Stimulation for Metabolic Dysregulation in Schizophrenia

Jianhua Chen
2026-11-04 1 min 61 words TDCS TACS TRNS

Summary: Individuals with schizophrenia (SCZ) frequently exhibit glycolipid metabolic disorders. This clinical issue cannot be fully explained by the high metabolic risk associated with antipsychotic medications alone. It is increasingly recognized to involve complex multi-system interactions, particularly t...

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Effects of Different Stimulation Parameters on the Efficacy of Transcranial Direct Current Stimulation for Post-Stroke Dysphagia: An Updated Systematic Review and Meta-Analysis of Randomized Controlled Trials

👤 Yuan Lin
📅 2026-11-04 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS yields favorable effects on swallowing function in patients with PSD when administered to the lesional hemisphere, the contralesional hemisphere, or bilaterally, and there are no differences in efficacy among the three stimulation modes. An intensity of 1.8 mA represents a potential...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

Jana Zweerings 2026-11-04 1 min 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-11-05 1 min 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

Jing-Ning Zhu 2026-11-05 1 min 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti
📅 2026-11-05 📖 1 min 📝 41 words

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Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

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Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent Prefrontal Neurochemical Effects of High-Definition Transcranial Direct Current Stimulation: Directional E-field Associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory 📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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